

Book.csv (Dataset)

ex.	sky	Airtemp	Humidity	Wind	Water	Forecast	Enjoy
1	sunny	warm	Normal	strong	warm	same	yes
2	sunny	warm	high	strong	warm	same	yes
3	rainy	cold	high	strong	warm	same	no
4	sunny	warm	high	strong	warm	same	yes

output:

['Sunny', 'warm', 'Normal', 'strong', 'warm', 'same', 'yes']
 ['Sunny', 'warm', 'high', 'strong', 'warm', 'same', 'yes']
 ['rainy', 'cold', 'high', 'strong', 'warm', 'same', 'No']
 ['Sunny', 'warm', 'high', 'strong', 'warm', 'same', 'yes']

6

The most specific hypothesis is: ['0', '0', '0', '0', '0', '0']
 The most generic hypothesis is: ['?', '?', '?', '?', '?', '?']

Experiment No. 01 Date 12/02/25 Page No 01
 Name of Experiment Find-S Algorithm

Lab Program-01

1) Implement and Demonstrate the find-S algorithm for finding the most specific hypothesis based on given set of training data sample.
 Read the training data from .csv file.

*) Import csv

a = []

with open('Book.csv') as cfile:

reader = csv.reader(cfile)

for row in reader:

a.append(row)

print(row)

num-attribute = len(a[0]) - 1

print(num-attribute)

*) hypothesis = a[0][:-1]

print(hypothesis)

*) Print ("the most specific hypothesis:", ["0"] * num-attribute)

print ("the most generic hypothesis:", ["?"] * num-attribute)

['sunny', 'warm', 'normal', 'strong', 'warm', 'same']

Find S: Finding a maximally specific hypothesis is

The training example no:1 The hypothesis is

['sunny', 'warm', 'normal', 'strong', 'warm', 'same']

The training example no:2 The hypothesis is

['sunny', 'warm', 'p', 'strong', 'warm', 'same']

The training example no:3 The hypothesis is

['sunny', 'warm', 'p', 'strong', 'warm', 'same']

The training example no:4 The hypothesis is

['sunny', 'warm', 'p', 'strong', 'p', 'p']

The maximally specific hypothesis for the training

['sunny', 'warm', 'p', 'strong', 'p', 'p']

Experiment No.....01..... Date.....12/02/2025..... Page No.....
Name of Experiment.....Find S: Algorithm.....

*) hypothesis = a[0][: -1]

print(hypothesis)

print("\n Find S: Finding a maximally

specific hypothesis")

for i in range(len(a)):

if a[i][num-attribute] == "Yes":

for j in range(num-attribute):

if a[i][j] != hypothesis[j]:

hypothesis[j] = 'p'

print("\nThe training example no:", i+1,
"the hypothesis is:", hypothesis)

print("\n the maximally specific hypothesis
for training set is")
print(hypothesis)